Aria Halavati

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Employment

Università Bocconi — Milan, Italy
Postdoc, funded by ERC Project ANGEVA — Sep 2025–Present
- Mentor: Antonio De Rosa

Education

Courant Institute of Mathematical Sciences, New York University — New York, NY
Ph.D. in Pure Mathematics — Sep 2020–May 2025
- Advisors: Guido De Philippis and Fang-Hua Lin
- Research area: Geometric measure theory and partial differential equations

Department of Mathematics, Sharif University of Technology — Tehran, Iran
B.S. in Mathematics — 2015–2019
- Advisor: Morteza Fotouhi

Publications and Preprints

1. **The anisotropic Weyl law and Grassmannian equidistribution**.
   Antonio De Rosa, Aria Halavati. Preprint (2026), arXiv:2610.18599.
2. **On Cylindrical singularities of minimal hypersurfaces**.
   Aria Halavati, Luca Spolaor. Preprint (2026), arXiv:2610.10754.
3. **Integral curvature estimates and geodesic limits for stable abelian Higgs fields**.
   Marco Badran, Marco A. M. Guaraco, Aria Halavati. Preprint (2026), arXiv:2609.21893.
4. **Min-Max construction of anisotropic minimal surfaces with genus bound**.
   Antonio De Rosa, Aria Halavati, Ling Wang. Preprint (2026), arXiv:2606.30098.
5. **Carleman estimates for stationary Q-valued maps: a variational approach**.
   Aria Halavati, Luca Spolaor. Preprint (2025), arXiv:2508.14388.
6. **Decay of excess for the abelian Higgs model**.
   Guido De Philippis, Aria Halavati, Alessandro Pigati. *J. Eur. Math. Soc.*, Online First (2026). doi:10.4171/JEMS/1819.
7. **Quantitative stability of Yang–Mills–Higgs instantons in two dimensions**.
   Aria Halavati. *Arch. Ration. Mech. Anal.* **248**, Article 88 (2024). doi:10.1007/s00205-024-02035-2.
8. **New weighted inequalities on two-manifolds**.
   Aria Halavati. *Rend. Sem. Mat. Univ. Padova*, Online First (2026). doi:10.4171/RSMUP/206.

Invited Talks

Westlake Winter School in Geometric Measure Theory | Westlake University, Hangzhou, China; invited individual lecture (upcoming) — 18–22 Jan 2027
Oberwolfach Workshop on Calculus of Variations | Mathematisches Forschungsinstitut Oberwolfach, Germany; speaker — 30 Aug–4 Sep 2026
Scuola Normale Superiore di Pisa | Pisa, Italy — 5–7 Nov 2025
UNAM | Juriquilla, Querétaro, Mexico — 6–8 Aug 2025
cSplash, Courant Institute | New York, NY — 5 Apr 2025
UT Austin Analysis Seminar | Austin, TX — 12 Mar 2025
Princeton DGGA Seminar | Princeton, NJ — 26 Feb 2025
MIT Analysis Seminar | Boston, MA — 12 Nov 2024
UC San Diego Geometry Seminar | San Diego, CA — 31 Oct 2024

Invited Talks (continued)

Stanford Geometry Seminar | Stanford, CA — 16 Oct 2024
Virtual Analysis and PDE Seminar, hosted by UC Irvine | Online — 3 Oct 2024
Cornell Analysis Seminar | Ithaca, NY — 9 Sep 2024
Courant Geometric Analysis Seminar | New York, NY — 6 Sep 2024
University of Toronto Student PDE Seminar | Online — 14 Feb 2024

Workshops and Conferences

International geometric analysis conference in Milan II | Bocconi University, Milan, Italy — 8–12 Jun 2026
SLMath – Geometry and Analysis of Special Structures on Manifolds | Berkeley, CA — 18–22 Nov 2024
SLMath – Recent Progress on Geometric Analysis and Riemannian Geometry | Berkeley, CA — 21–25 Oct 2024
Joint Mathematics Meetings 2024 | San Francisco, CA — 3–6 Jan 2024
Oberwolfach PDE Workshop 2330 (Invited) | Germany — 23–29 Jul 2023

Awards and Honors

Peter Lax Fellowship, awarded to an outstanding Ph.D. student, Courant Institute — 2022
First Prize, International Mathematical Competition, UCL — 2017
Gold Medal, International Mathematical Olympiad — 2015
Gold Medal, Iran's National Mathematical Olympiad — 2014

Academic Service

Reviewer for: *Canadian Journal of Mathematics*; *Journal of Functional Analysis*; *Annales de l'Institut Henri Poincaré, Analyse Non Linéaire*; *Adv. Math.*

Co-organized: Courant Graduate Student/Postdoc Seminar; social chapter of the Courant student organization; social climate survey administered for Courant student feedback; geometric measure theory reading group at Courant.

Teaching

SLMath, Berkeley — 17–28 Jul 2024
Teaching Assistant for the summer school *Special Geometric Structures and Analysis*, St. Mary's College
- An introduction to geometric measure theory.

Courant Institute of Mathematical Sciences, NYU. Teaching Assistant.

Calculus I	Fall 2019; Fall 2021
Math for Econ	Spring 2020
Great Ideas in Math for non-math majors	Spring 2020
Linear Algebra	Fall 2022; Spring 2025
Probability and Statistics	Spring 2022
Analysis	Fall 2023; Spring 2024; Fall 2024

Young Scholars Club, Tehran, Iran. Tutoring Iran's IMO team — 2016
Kerman, Iran. Math classes for students in my hometown; 7 national and 1 IMO medal from my class — 2014–2015

Skills

Languages: Persian, English.
Programming: Python, TensorFlow, PyTorch, C++, MATLAB.
Music: Piano, Setar, Guitar, Oud.